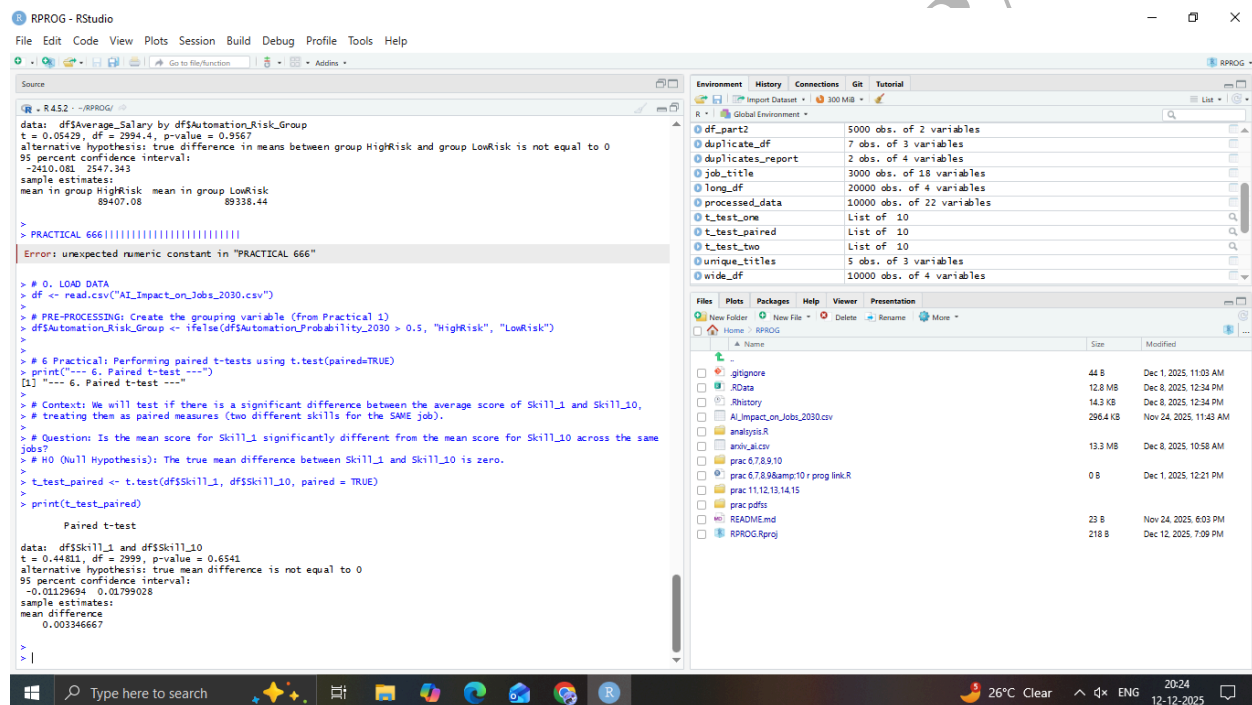


SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME : DATA ANALYSIS WITH SAS/SPSS/R

PRACTICAL 6

AIM: Performing paired t-test using t.test(paired=TRUE)



The screenshot displays the RStudio interface. The Source pane on the left contains R code for loading data, creating a grouping variable, and performing a paired t-test. The Environment pane on the right shows the objects created in the workspace.

```
> R 4.3.2 - ~/RPROG/
data: df$Average_Salary by df$Automation_Risk_Group
t = 0.05429, df = 2994.4, p-value = 0.9567
alternative hypothesis: true difference in means between group HighRisk and group LowRisk is not equal to 0
95 percent confidence interval:
-2410.081 2547.343
sample estimates:
mean in group HighRisk mean in group LowRisk
89407.08 89338.44

> PRACTICAL 666|
Error: unexpected numeric constant in "PRACTICAL 666"

> # 0. LOAD DATA
> df <- read.csv("AI_Impact_on_Jobs_2030.csv")
>
> # PRE-PROCESSING: Create the grouping variable (from Practical 1)
> df$Automation_Risk_Group <- ifelse(df$Automation_Probability_2030 > 0.5, "HighRisk", "LowRisk")
>
> # 6 Practical: Performing paired t-tests using t.test(paired=TRUE)
> print("---- 6. Paired t-test ----")
[1] "---- 6. Paired t-test ----"
>
> # Context: We will test if there is a significant difference between the average score of Skill_1 and Skill_10,
> # treating them as paired measures (two different skills for the SAME job).
>
> # Question: Is the mean score for Skill_1 significantly different from the mean score for Skill_10 across the same
> # jobs?
> # H0 (Null Hypothesis): The true mean difference between Skill_1 and Skill_10 is zero.
>
> t_test_paired <- t.test(df$Skill_1, df$Skill_10, paired = TRUE)
> print(t_test_paired)

Paired t-test
data: df$Skill_1 and df$Skill_10
t = 0.44811, df = 2999, p-value = 0.6541
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
-0.01129694 0.01799028
sample estimates:
mean difference
0.00346667

>
> |
```

Object	Class	Attributes
df_part2	data.frame	5000 obs. of 2 variables
duplicate_df	data.frame	7 obs. of 3 variables
duplicates_report	data.frame	2 obs. of 4 variables
job_title	data.frame	3000 obs. of 18 variables
long_df	data.frame	20000 obs. of 4 variables
processed_data	data.frame	10000 obs. of 22 variables
t_test_one	list	List of 10
t_test_paired	list	List of 10
t_test_two	list	List of 10
unique_titles	data.frame	5 obs. of 3 variables
wide_df	data.frame	10000 obs. of 4 variables

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S086